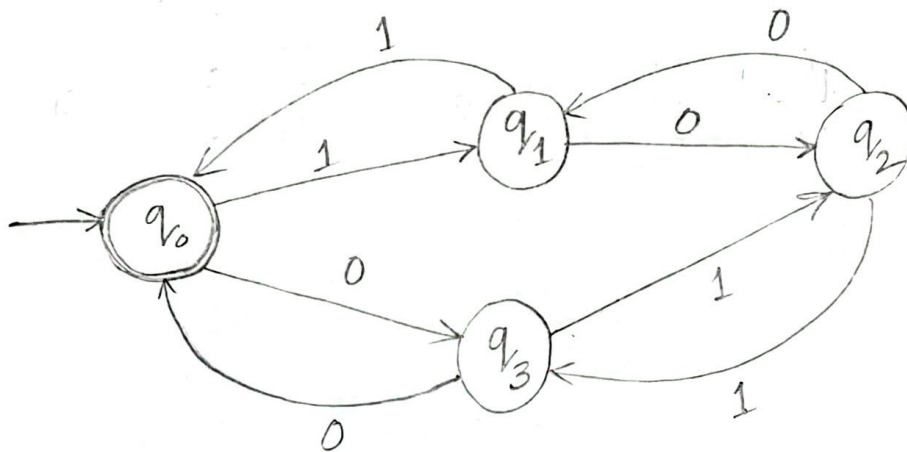


(a)

Has both even number of 0's and even number of 1's.

DFA:

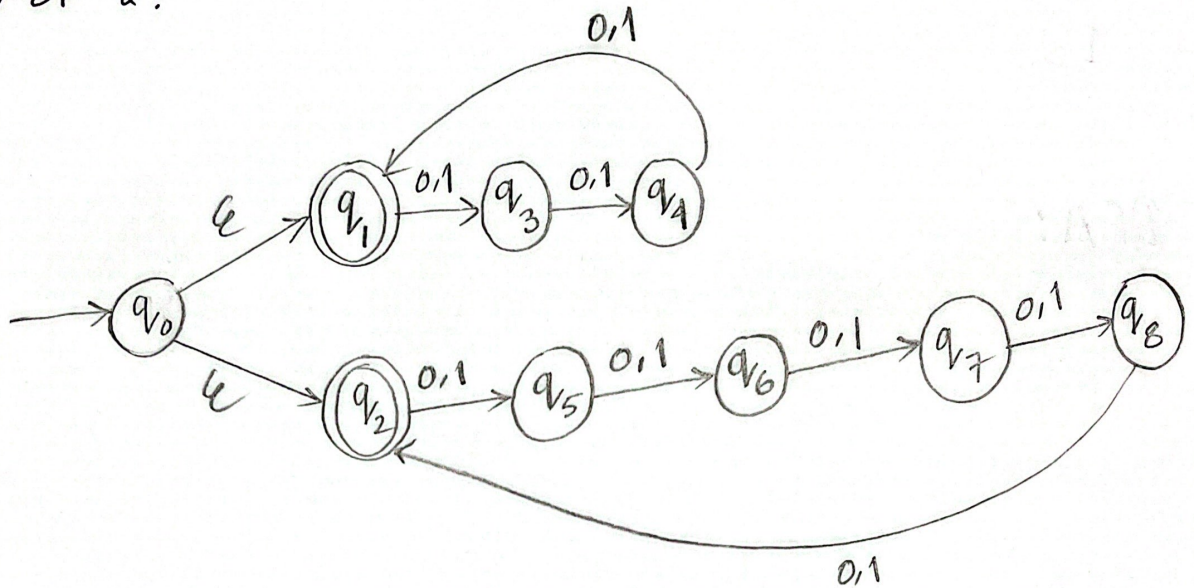


NFA: For this case, NFA and DFA are both the same.

(b)

Binary representation of the strings is divisible by 3 or 5.

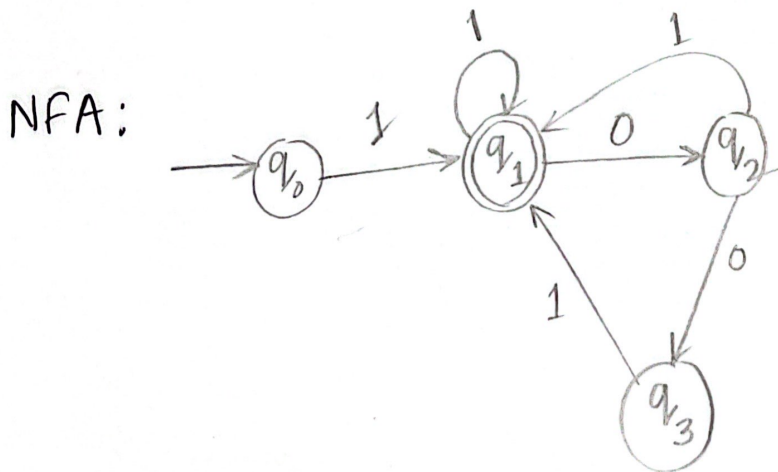
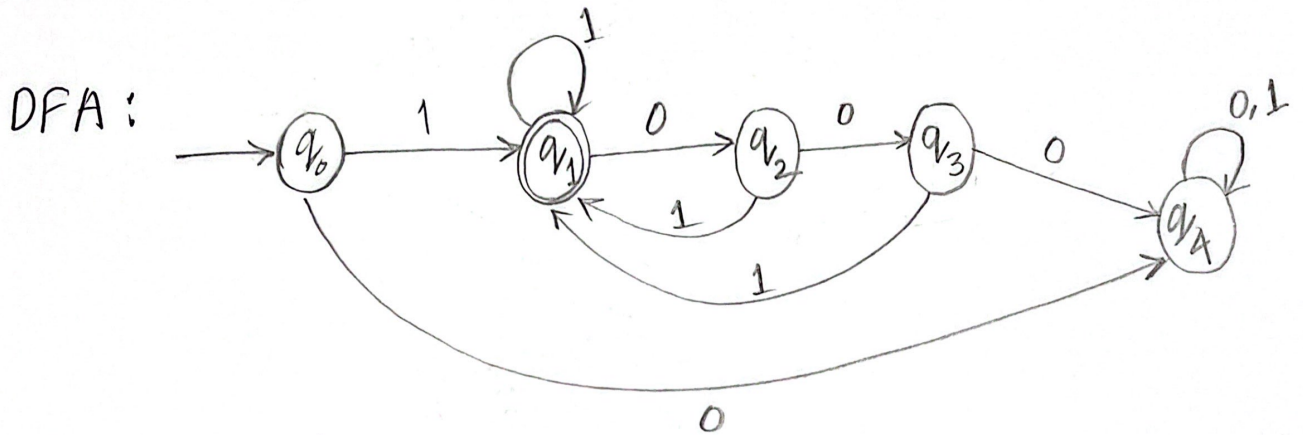
DFA:



NFA: For this case, NFA and DFA are both same.

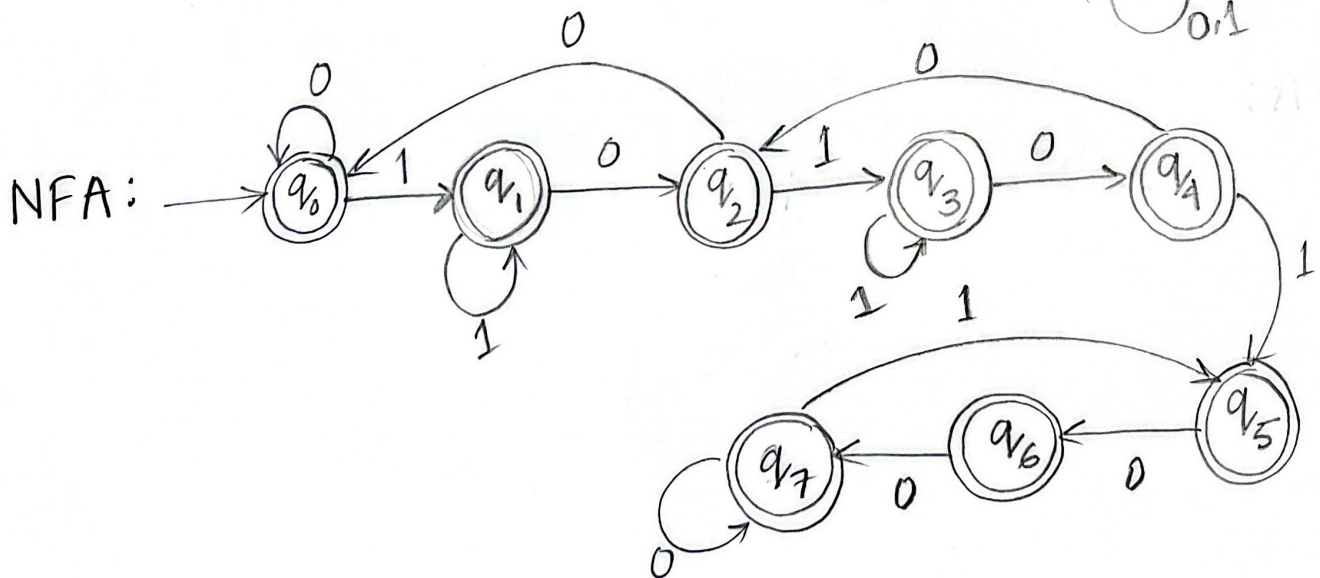
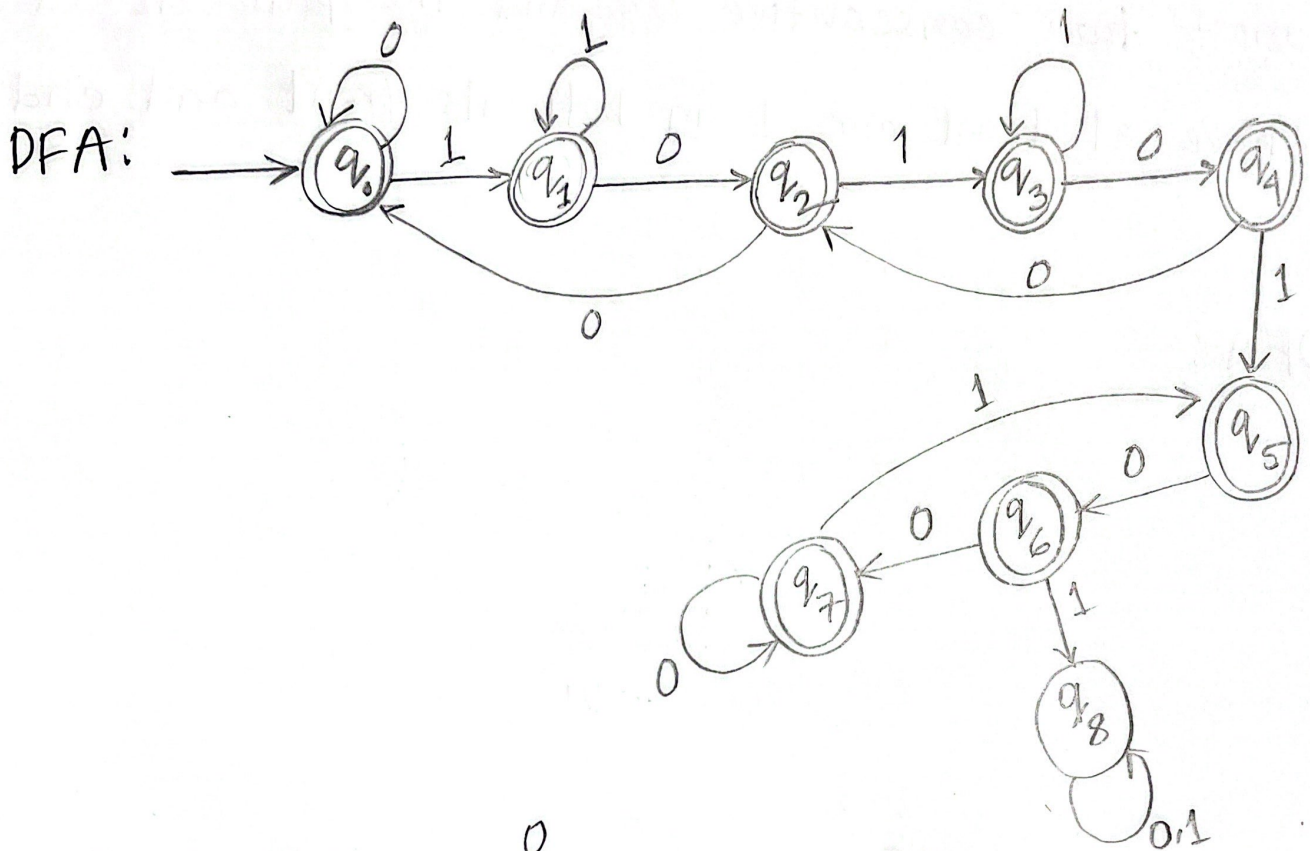
(c)

Non-empty string such that there can be at most two consecutive 0's, and the pair should have at least one 1 in both its front and end.



(d)

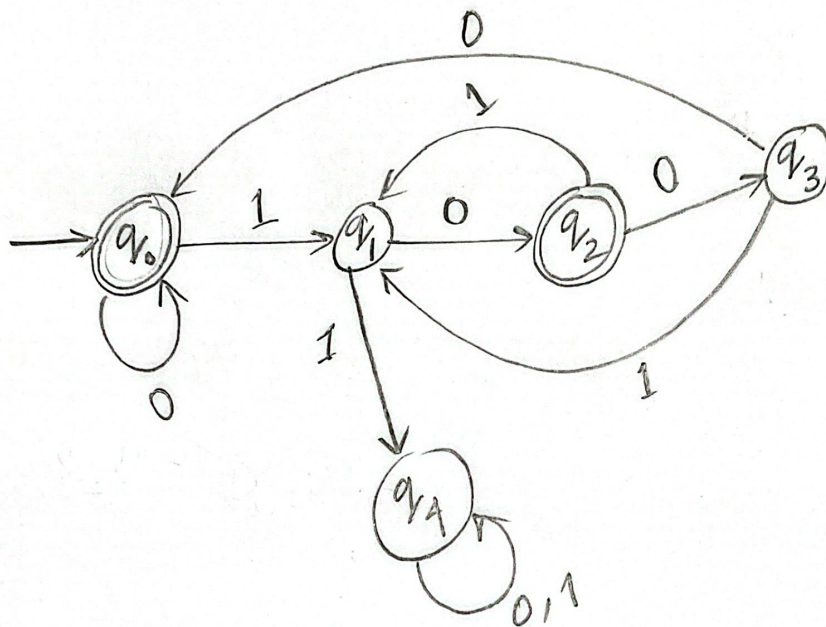
Contains substring 101 at most two times



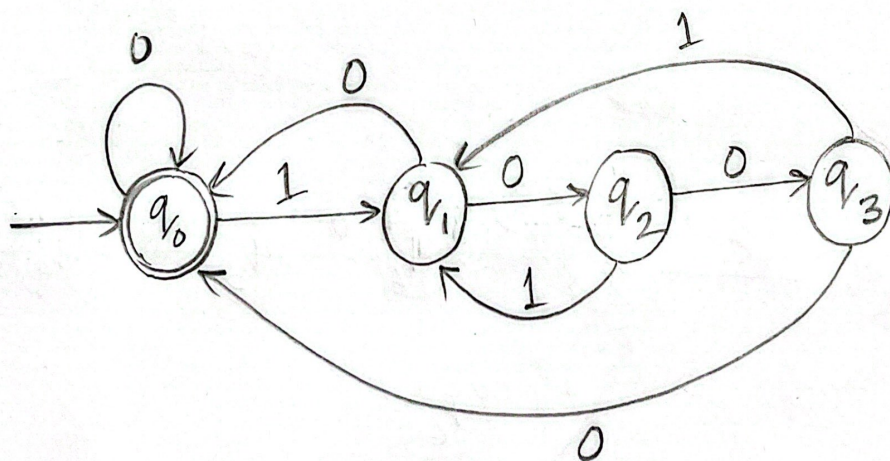
(e)

Each 1 is followed by at least one 0 but cannot end with 100.

DFA:



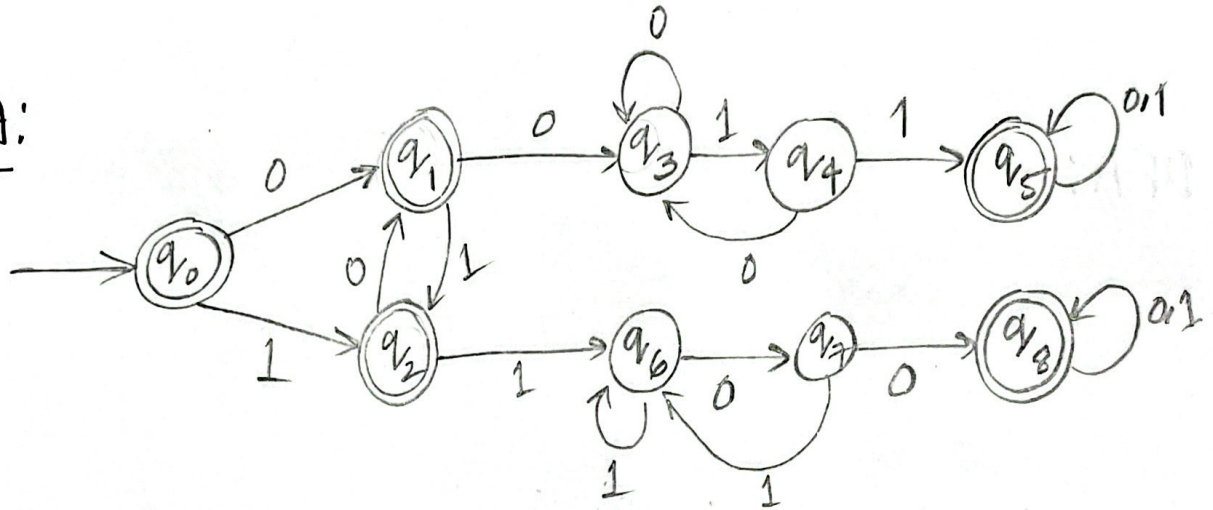
NFA:



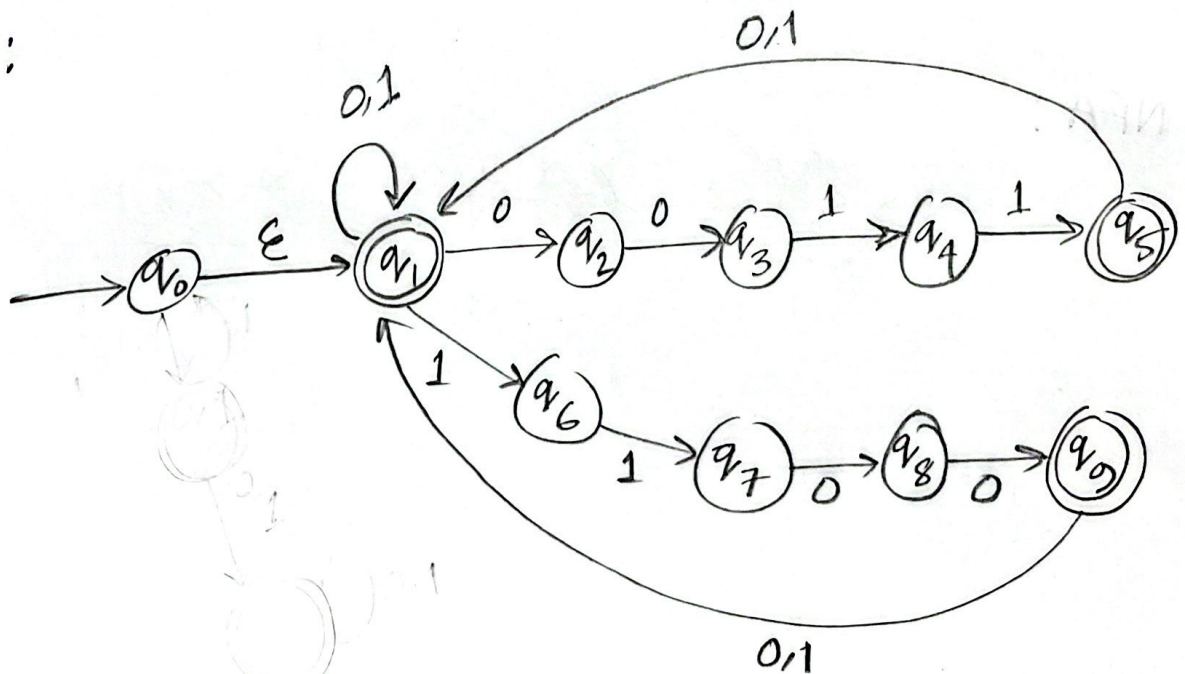
(f)

When substring 00 is found, substring 11 should exist as well and vice versa.

DFA:



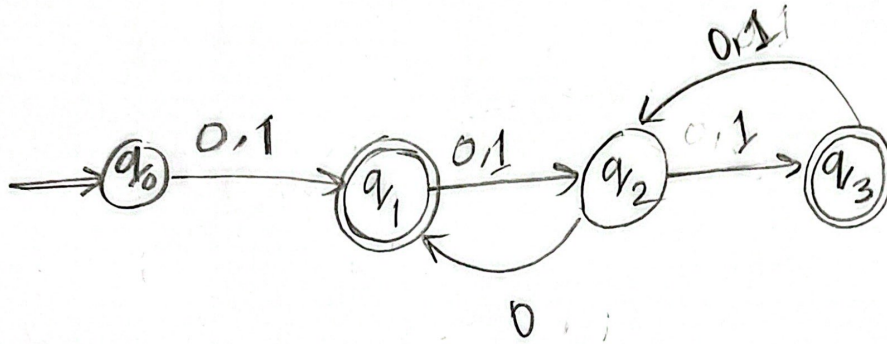
NFA:



(9)

The absolute value of the difference between the number of 0's and the number of 1's is odd.

DFA:

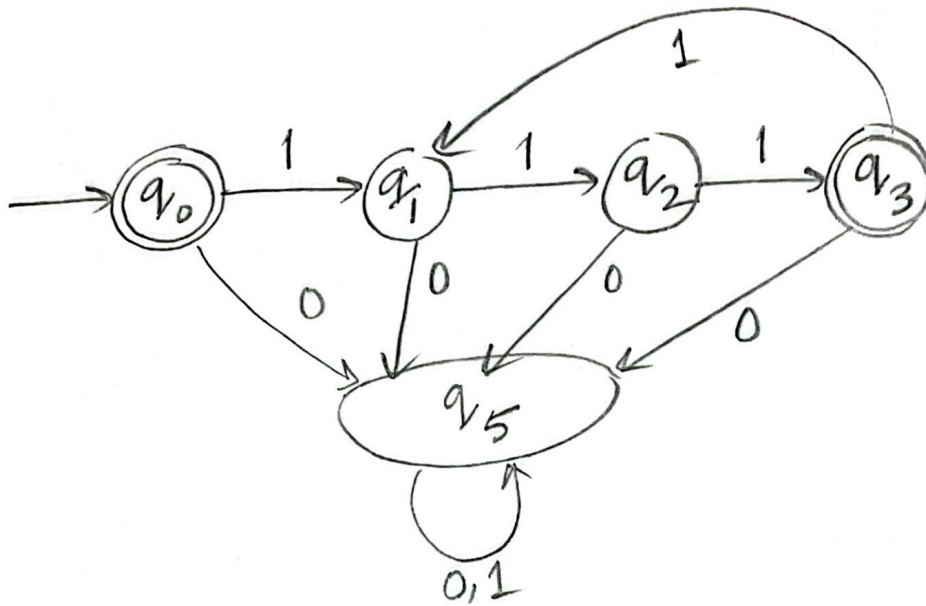


NFA: For this case, NFA and DFA are both the same.

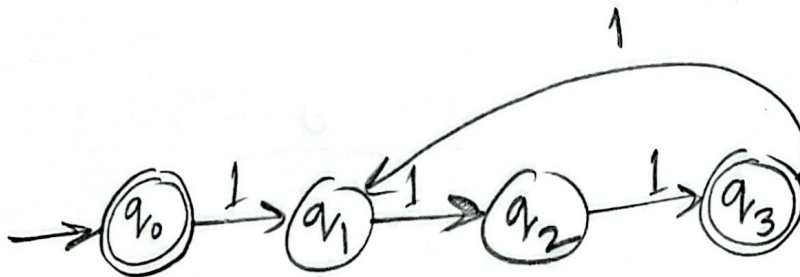
Ans. to the ques. no. 2

(a) $L_1 = \{w \mid w = 1^n, n \text{ is a multiple of } 3\}$

DFA:

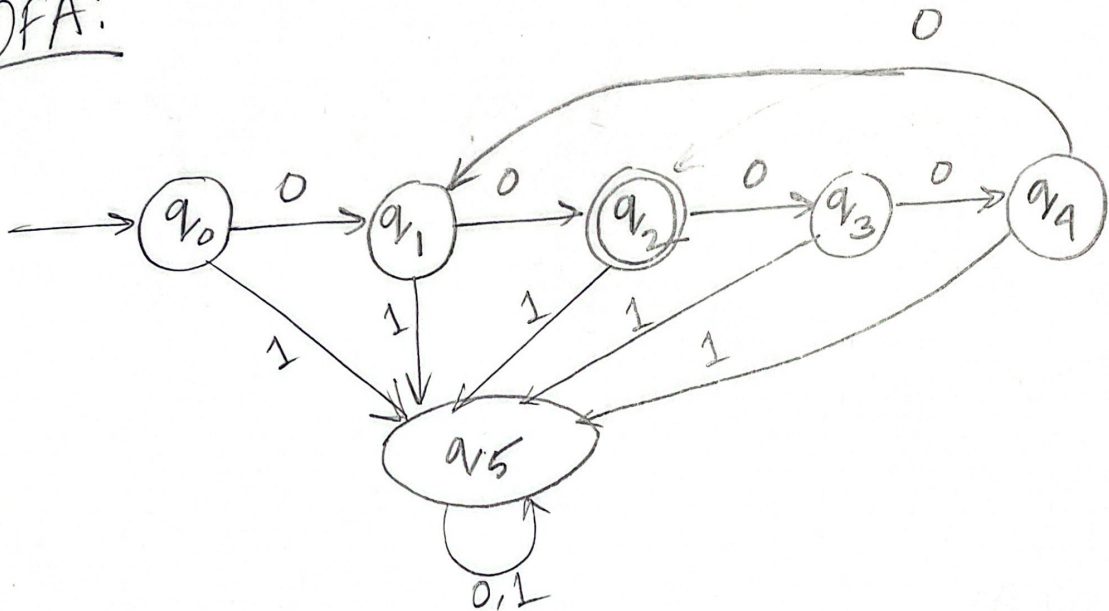


~~(b)~~ NFA:

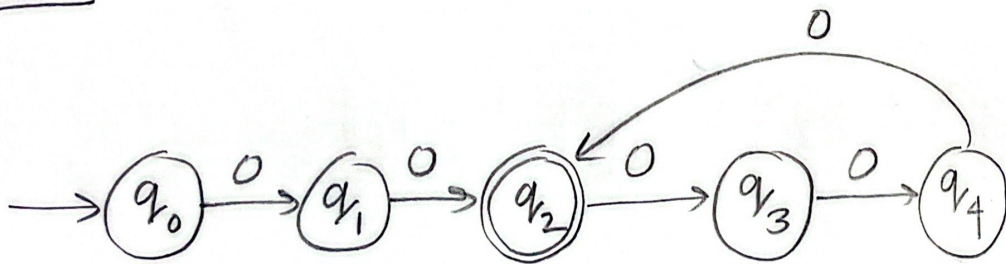


(b) $L_2 = \{w \mid w = 0^m, m \text{ is 2 more than multiple of 4}\}$

DFA:

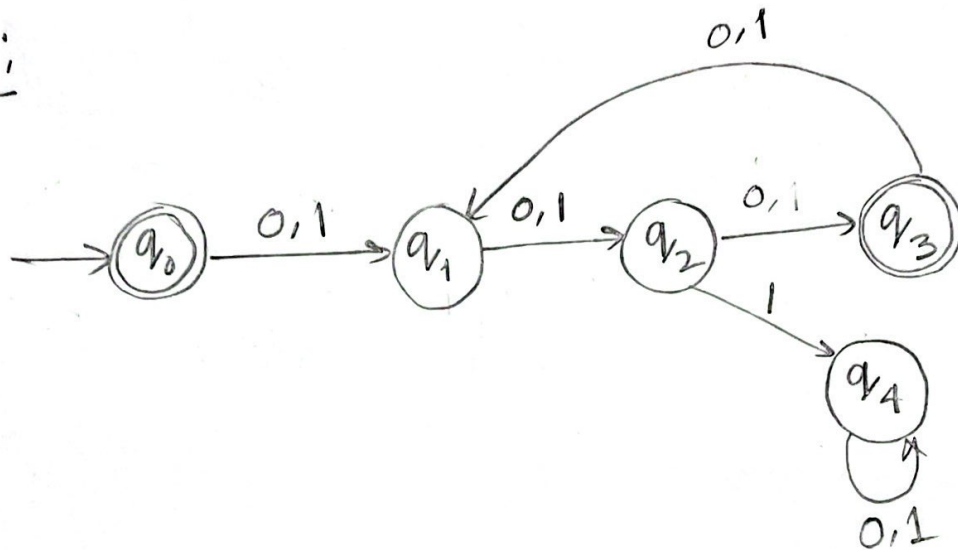


NFA:

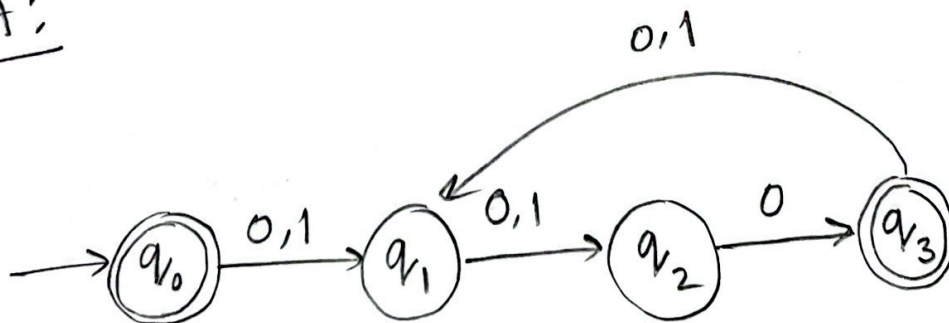


(c) $L_3 = \{w \mid w \text{ contains 0 in every position which is multiple of 3}\}$

DFA:

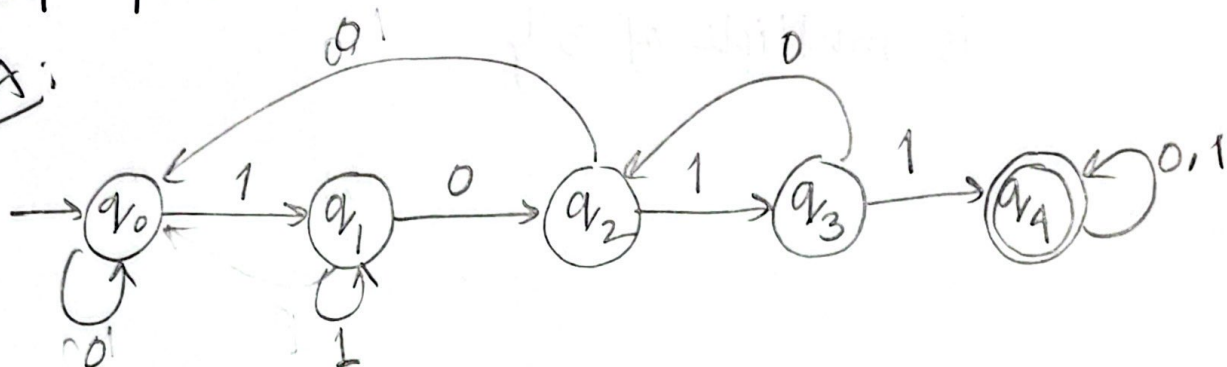


NFA:

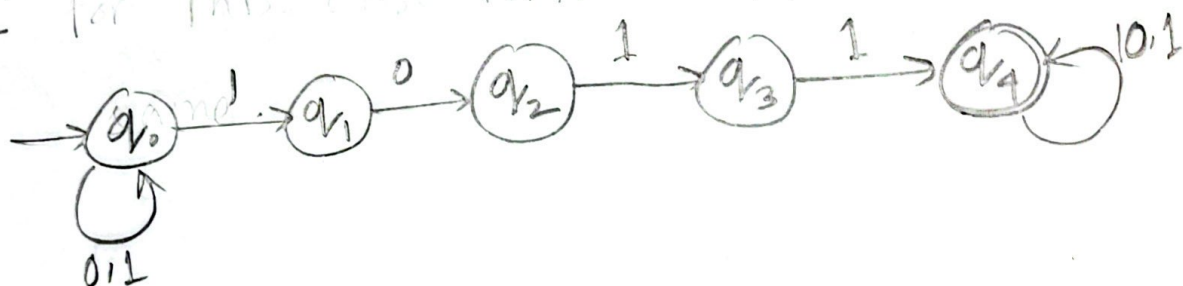


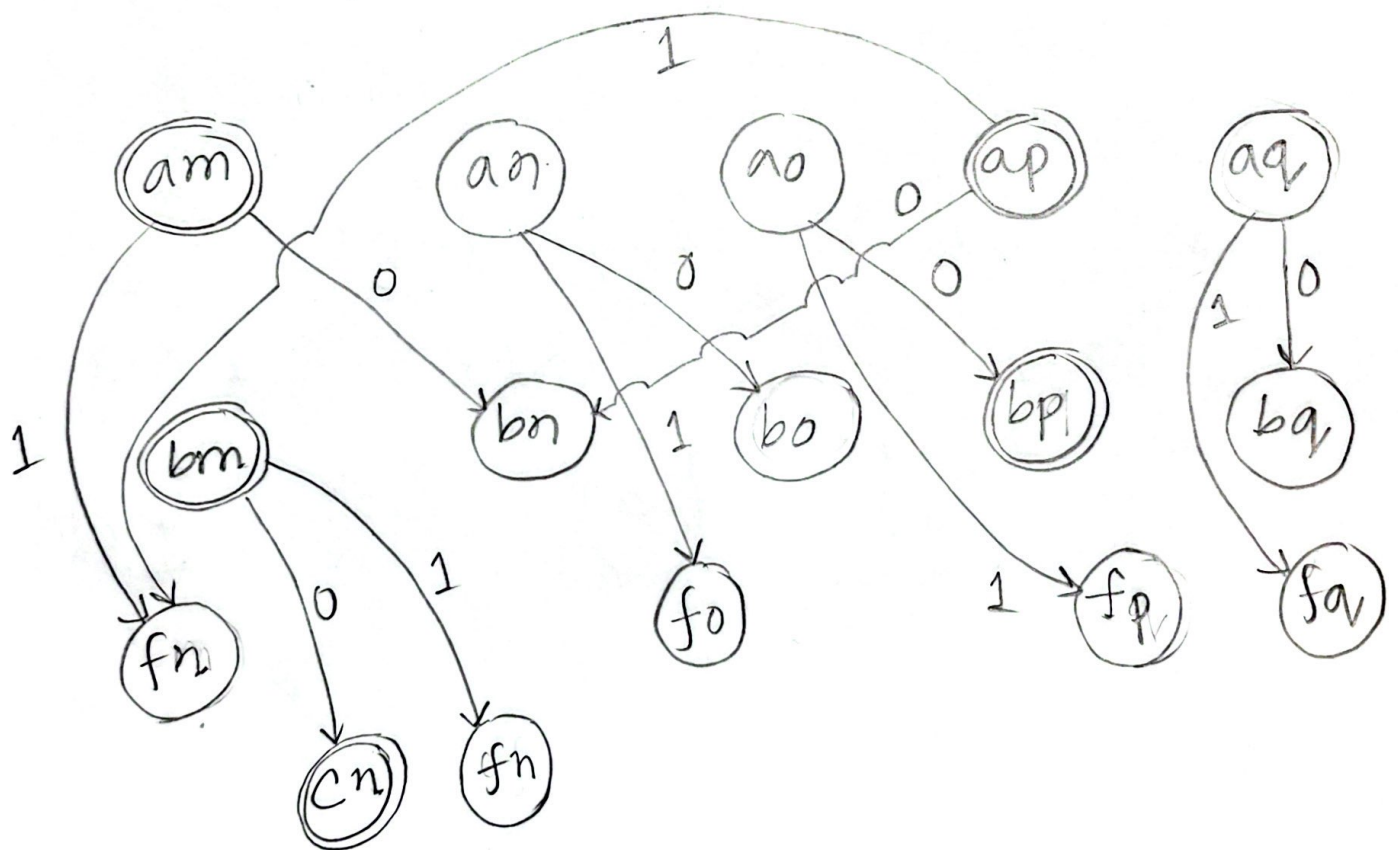
(d) $L_4 = \{w \mid w \text{ contains substring } 1011\}$

DFA:



NFA: For this case NFA and DFA are the



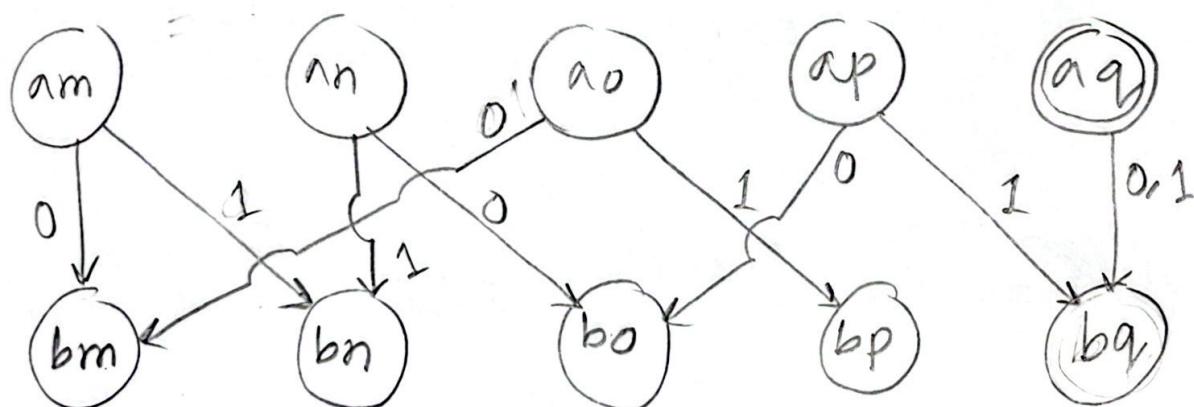
$$L_2 \cup L_3$$
$$L_2 = \{ \overset{q_0}{a}, \overset{q_1}{b}, \overset{q_2}{c}, \overset{q_3}{d}, \overset{q_4}{e}, \overset{q_5}{f} \}$$
$$L_3 = \{ m, n, o, p, q \}$$
$$L_2 \times L_3 = \{am, an, ao, ap, aq, bm, bn, bo, bp, bq, \\ cm, cn, co, cp, cq, dm, dn, do, dp, dq, \\ em, en, eo, ep, eq, fm, fn, fo, fp, fq\}$$


$L_3 \cap L_4$

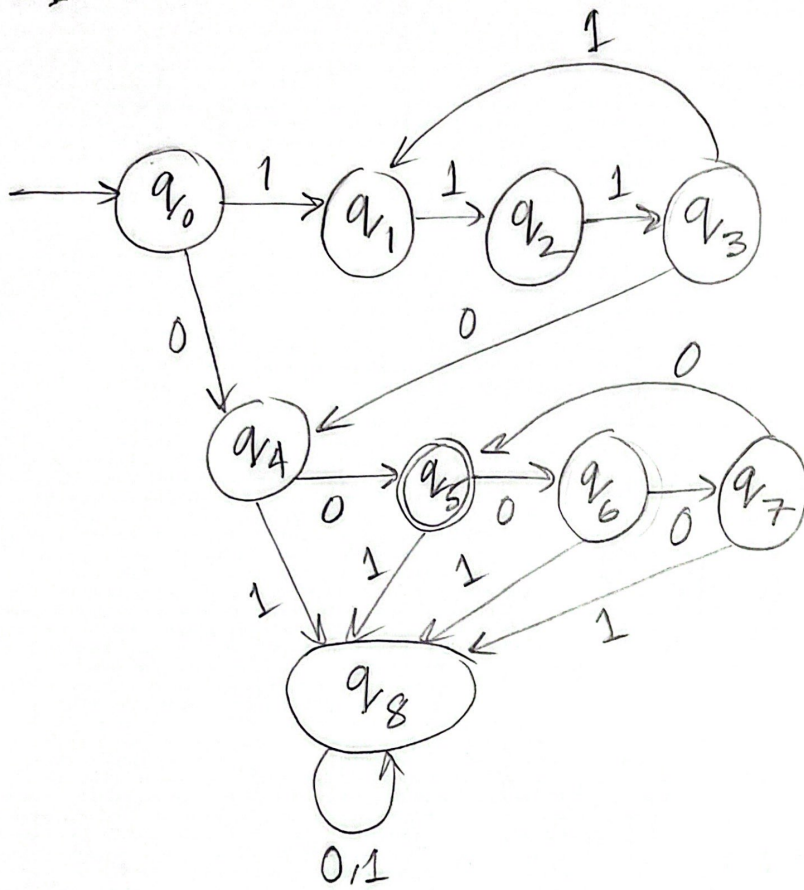
$L_3 = \{ \overset{q_0}{a}, \overset{q_1}{b}, \overset{q_2}{c}, \overset{q_3}{d}, \overset{q_4}{e} \}$

$L_4 = \{ \overset{q_0}{m}, \overset{q_1}{n}, \overset{q_2}{o}, \overset{q_3}{p}, \overset{q_4}{q} \}$

$L_3 \times L_4 = \{ am, an, ao, ap, aq, bm, bn, bo, bp, bq, cm, cn, co, cp, cq, dm, dn, do, dp, dq, em, en, eo, ep, eq \}$



$L_1 \cdot L_2$

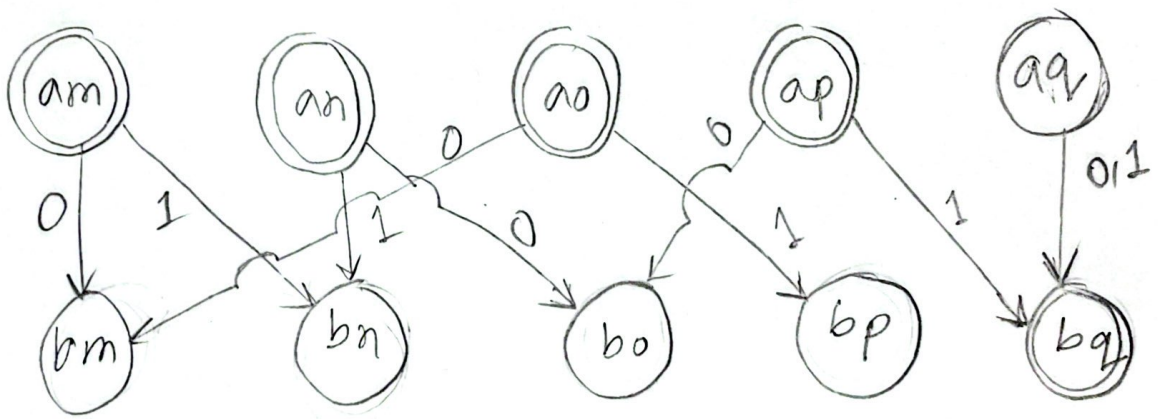


$$L_3 \oplus L_4$$

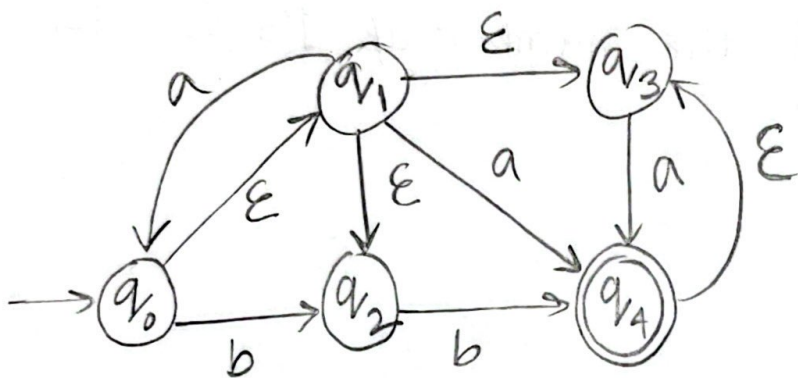
$$L_3 = \{a, b, c, d, e\}$$

$$L_4 = \{m, n, o, p, q\}$$

$$L_3 \times L_4 = \{am, an, ao, ap, aq, bm, bn, bo, bp, bq, \\ cm, cn, co, cp, cq, dm, dn, do, dp, dq, \\ em, en, eo, ep, eq\}$$



Ans. to the ques. no. 3



NFA	a	b
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_2, q_3, q_4\}$
$\{q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_3, q_4\}$
$\{q_2\}$	$\emptyset$	$\{q_3, q_4\}$
$\{q_3\}$	$\{q_3, q_4\}$	$\emptyset$
$\{q_3, q_4\}$	$\{q_3, q_4\}$	$\emptyset$
$\{q_4\}$	$\emptyset$	$\emptyset$

NFA	a	b
$\rightarrow \{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_2, q_3, q_4\}$
$\{q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_3, q_4\}$
$\{q_2\}$	$\emptyset$	$\{q_3, q_4\}$
$\{q_3\}$	$\{q_3, q_4\}$	$\emptyset$
$\{q_3, q_4\}$	$\{q_3, q_4\}$	$\emptyset$
$\{q_4\}$	$\emptyset$	$\emptyset$

DFA	a	b
$\rightarrow \{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_2, q_3, q_4\}$
* $\{q_2, q_3, q_4\}$	$\{q_3, q_4\}$	$\{q_3, q_4\}$
$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_0, q_1, q_2, q_3, q_4\}$	$\{q_2, q_3, q_4\}$
$\{q_3, q_4\}$	$\{q_3, q_4\}$	Trap

